

Examiner
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7. Fish farming is an important industry in Scotland, producing almost 180 000 tonnes of fish in 2014. It provides employment and financial gain for the areas where it is carried out. The number and size of fish farms has increased as one way of preventing overfishing.

The table below shows the estimated use and loss of two elements during the production of fish in Scotland by fish farming.

Nutrient output per tonne of production	Nitrogen	Phosphorus
used in feed (kg)	86	18
lost as uneaten food (kg)	4	1
lost as organic matter in faeces (kg)	12	3
lost as inorganic products (e.g. ammonium ions and phosphate ions) (kg)	37	9

Adapted from “The Interaction between Fish Farming and Algal Communities of the Scottish Waters – A Review”, Rydberg, Sjöberg and Stigebrandt. Research report 2003/04.

With reference to the information provided in the table and your own knowledge, explain the possible ecological impacts of fish farming in the areas where it takes place. Describe other ways to overcome the problem of overfishing. Explain how these methods are used to prevent overfishing. [9 QER]

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